

Tools for Applied Monitoring, Evaluation and Learning

Beyond Analysis: Strategic Evaluation with Stata

**Modern Workflows for Implementation Science and
Embedded Research**

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Preface: The Modern Evaluator's Dilemma

Applied evaluation has moved beyond simple pre-post t-tests. Today's evaluator must be part data engineer, part AI prompt engineer, and part visualization specialist. This book provides a roadmap for using Stata as the central "brain" of this complex ecosystem, focusing on the needs of practitioners working in high-stakes, resource-constrained environments.

**PART I: THE EMBEDDED
EVALUATOR
PRINCIPLES, SPEED, AND THE
MODERN STACK**

Strategic Evaluation in Implementation Science

1

1.1 The Practitioner-Researcher Relationship

Explores the "Embedded Evaluation" model. We discuss moving away from "parachute research" toward sustained relationships where the evaluator is a strategic partner in the implementation team.

- ▶ **Bridging the gap:** Shared goals vs. analytical independence. How to maintain rigor while being "in the room" where decisions are made.
- ▶ **Feedback loops:** Using Stata to provide real-time insights for program adaptation.
- ▶ **Messy Contexts:** Working with nonprofits where data collection is often secondary to service delivery.

1.1.1 Managing Power Dynamics in Embedded Evaluation

Strategies for navigating data ownership and the "bad news" problem when evaluation findings conflict with program goals.

1.1.2 Building Evaluation Literacy with Staff

Using Stata-generated data dictionaries and simple summaries to help non-technical staff understand how their inputs affect the final analysis.

Ado-file Idea: `evalaudit`

Proposed Program: `evalaudit`. This tool would generate a structured report auditing data-sharing agreements, quasi-identifiers, and reporting timelines to ensure compliance with funder mandates and ethical standards from day one. test

1.2 Implementation Science Principles in Practice

Implementation science focuses on the 'how' and 'why' of program success, not just the 'what'. See [James2013] for foundational concepts.

This section defines the core IS frameworks (e.g., RE-AIM, CFIR) and how they translate to data structures in Stata.

- ▶ **Fidelity Monitoring:** Tracking if the program is being delivered as intended.
- ▶ **Adaptation Tracking:** Coding and managing changes made to the intervention in real-time.

1.2.1 Iterative Evaluation: The PDSA Cycle in Stata

How to structure do-files to handle the Plan-Do-Study-Act cycle, where data is analyzed and results are used to adjust implementation in the next month.

1.2.2 Contextualizing the Evidence: Using CFIR to Code Site Characteristics

Using qualitative coding and site-level variables to explain why an intervention worked in one school but failed in another.

Ado-file Idea: **fidplot**

Proposed Program: `fidplot`. A specialized graphing tool for fidelity metrics that automatically compares observed implementation scores against a pre-set 'gold standard' or logic model benchmark.

1.3 Stata as the Central Evaluation Brain

Why Stata is the ideal bridge: its ability to handle messy, real-world data while providing the rigor required for publication and the speed required for stakeholders.

1.3.1 Bypassing Excel: The Risky Habit of Spreadsheet Analysis

Identifying common errors in spreadsheet-based evaluation and providing a workflow for transitioning nonprofits to code-based reproducibility.

Ado-file Idea: **projsetup**

Proposed Program: `projsetup`. A command to initialize the standardized directory structure, global macro hubs, and master do-files described in this book, ensuring a consistent environment for team-based evaluation.

Setting Up a High-Performance Environment

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2.1 The Speed Revolution: ftools and gtools

Detailed guide on using faster alternatives to core commands.

- **Benchmarking:** Comparing gcollapse vs. collapse on large administrative datasets.
- **Multi-way Fixed Effects:** Using reghdfe for complex nested evaluation designs.

When working with millions of administrative records, standard Stata commands can become bottlenecks. We implement ftools and gtools to reclaim your time.

2.1.1 Large Data Workflows without MP

Tips for managing memory and optimizing sort/merge operations on standard Stata flavors when working with extensive health or education records.

Ado-file Idea: gtools_audit

Proposed Program: gtools_audit. This tool would scan an existing do-file and identify bottlenecks where standard commands could be replaced by gtools or ftools equivalents, projecting potential time savings.

2.2 Monitoring Real-Time Participation

Techniques for building "Command Centers" in Stata to track response rates and program reach.

- **Response Rate Trackers:** Automating daily reports from Qualtrics/REDCap APIs.
- **Attrition Alerts:** Flagging participants at risk of dropping out before they do.

2.2.1 Early Warning Systems: Statistical Process Control (SPC)

Applying manufacturing-style control charts to program implementation to detect when fidelity or participation drops below acceptable levels.

Ado-file Idea: reachcheck

Proposed Program: reachcheck. A program that compares participation demographics against a target population (using Census or parent-file data) to calculate representative weights and identify under-served sub-populations in real-time.

2.3 Communicating with Funders and Stakeholders

Using Stata to automate the "Monday Morning Update" for nonprofit boards and foundation officers.

2.3.1 Automating the Appendix

Using `collect` to manage exhaustive reporting tables that funders require but practitioners rarely read, keeping the main report focused on strategic insights.

Ado-file Idea: `fundertable`

Proposed Program: `fundertable`. A wrapper for `collect` and `putexcel` that generates pre-formatted "Executive Summary" tables commonly requested by major philanthropic foundations.

2.4 Integrating AI: The Gemini CLI Bridge

The `gemini` package allows Stata to communicate directly with Google's LLMs. We'll cover API key management and shell security.

2.4.1 Prompt Engineering for Evaluators

Moving from generic "analyze this" prompts to structured "categorize these 5,000 progress notes based on the following implementation barriers" requests.

Ado-file Idea: `gemini_verify`

Proposed Program: `gemini_verify`. A tool that takes AI-coded text and compares it to a subset of human-coded "gold standard" data, calculating inter-rater reliability and accuracy metrics for the LLM's performance.

PART II: DATA ENGINEERING MODERNIZATION, CLEANING, AND HARMONIZATION

3.1 Web Scraping and API Pipelines

Automating data pulls from platforms using Stata.

3.1.1 OAuth and Secure API Handshakes

Managing credentials and security when pulling data from clinical or educational platforms directly into Stata's memory.

Ado-file Idea: `redcap_pull`

Proposed Program: `redcap_pull`. A simplified interface for common REDCap API calls that handles the heavy lifting of JSON parsing and variable labeling automatically.

3.2 Unstructured Data: The AI Bridge

Using LLMs within Stata to turn messy program notes into structured variables.

3.2.1 Sentiment Analysis as an Evaluation Tool

Extracting participant tone and satisfaction from open-ended survey responses to quantify "soft" implementation outcomes.

Ado-file Idea: `text2vars`

Proposed Program: `text2vars`. A program leveraging the Gemini API to scan unstructured progress reports and extract Key Performance Indicators (KPIs) into numeric variables.

Harmonization and Master Data Management

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4.1 Merging the Unmergable

Fuzzy matching techniques and building robust crosswalks.

4.1.1 Spatial Joining without GIS

Using Stata for proximity joins (e.g., matching schools to the nearest health clinic) using latitude/longitude data without specialized GIS software.

Ado-file Idea: fuzzymatch_pro

Proposed Program: fuzzymatch_pro. An extension of reclink that incorporates multiple string distance algorithms (Levenshtein, Jaro-Winkler) and generates a reviewable "Merge Quality Report" with suggestions for human review.

4.2 Longitudinal Panel Construction

Building "Long" files from "Wide" chaos in multi-year evaluations.

4.2.1 Transition Matrices: Visualizing Participant Movement

Creating datasets that track how participants move through different stages of a program (e.g., from 'Waitlisted' to 'Engaged' to 'Graduated').

Ado-file Idea: trackflow

Proposed Program: trackflow. A utility that generates Sankey-style data showing participant movement through program phases, ready for export to interactive web dashboards.

5.1 The Privacy-Utility Tradeoff

Techniques for de-identification and creating "Research-Ready" public-use files.

5.1.1 Measuring Re-identification Risk: k-Anonymity and l-Diversity

Implementing statistical metrics in Stata to calculate how likely a participant is to be re-identified based on their demographic profile.

Ado-file Idea: `riskscan`

Proposed Program: `riskscan`. A command that scans a dataset for potential Quasi-Identifiers and calculates re-identification risk metrics before a public-use file is released.

5.2 Synthetic Data Generation

Using Stata's `simulate` and AI-driven synthesis.

5.2.1 Validating the Fake: Statistical Fidelity Tests

How to prove to a client that your synthetic dataset maintains the same correlation structure as the real data without exposing the raw values.

Ado-file Idea: `synthgen`

Proposed Program: `synthgen`. A command that integrates with Python to generate synthetic cohorts that preserve non-linear relationships and interactions found in the original sensitive data.

**PART III: THE NEW CAUSAL
FRONTIER
INNOVATIVE AND
NON-STANDARD ANALYSIS**

6.1 The Revolution in Staggered Adoption

Explaining why Two-Way Fixed Effects (TWFE) often fails in applied settings.

6.1.1 The Forbidden Regression: Why Simple Interactions Aren't DiD

Identifying common pitfalls where evaluators incorrectly apply DiD logic to staggered treatment timing.

Ado-file Idea: `did_selector`

Proposed Program: `did_selector`. A diagnostic tool that analyzes treatment timing and suggest the most appropriate modern estimator (`csdid`, `jwdid`, or `did2s`) based on the data's structure.

6.2 Implementing `csdid` and `jwdid`

Step-by-step guide to estimating heterogeneous treatment effects.

Synthetic Controls and Matrix Completion

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7.1 The Synthetic Control Method (**synth**)

Building "counterfactual" units for niche evaluation targets.

7.1.1 In-space and In-time Placebo Tests

Validating synthetic control results through systematic falsification tests to gain stakeholder confidence.

7.2 Synthetic Difference-in-Differences (**sdid**)

Ado-file Idea: `sdid_viz`

Proposed Program: `sdid_viz`. A specialized plotting command that overlays the unit weights and counterfactual trends to make the complex 'Synthetic DiD' logic intuitive for non-experts.

Machine Learning for Evaluators

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8.1 Targeting Models with Lasso and Elastic Net

Identifying which participants are most likely to drop out or succeed.

8.1.1 Ethics of Prediction: Bias Detection in Targeting

Auditing ML models to ensure that program targeting isn't reinforcing existing systemic inequities.

Ado-file Idea: faircheck

Proposed Program: faircheck. A program that audits machine learning predictions across protected classes (race, gender, income) and reports potential bias in program targeting.

8.2 Random Forests in Stata

8.2.1 Variable Importance for Program Managers

Translating the black box of Random Forests into "Top 5" lists of factors that drive implementation success.

**PART IV: HIGH-IMPACT
COMMUNICATION
SPARKLINES, WEBDOCS, AND
INTERACTIVE REPORTING**

9.1 Sparklines and Micro-charts with sparkta

Using Fahad Mirza's package for high-density site reporting.

9.1.1 Maximizing the Data-to-Ink Ratio

Principles of accessible design for clinical and educational dashboards.

9.2 Advanced Coefficient Plots

Dynamic Reporting and Web Integration

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10.1 Dynamic HTML with webdoc

Building live, searchable project websites directly from your do-files.

10.1.1 The Self-Contained Project Portal

Workflows for embedding interactive maps and tables into a single HTML file that stakeholders can open without special software.

10.2 AI-Assisted Narrative Generation

Using the gemini package to draft the "Executive Summary."

10.2.1 The 'So What?' Test: Finding Meaning in the Table

Using AI to identify the single most impactful data point in a complex regression table for inclusion in a policy brief.

Ado-file Idea: stata2brief

Proposed Program: stata2brief. A command that takes current analysis results and a custom template to auto-generate a structured one-page summary brief via the Gemini API.

**PART V: BUILDING TOOLS
THAT Mata-ER
CUSTOM DEVELOPMENT AND
SCALABILITY**

Building Internal Evaluation Toolkits

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11.1 From Scripts to Systems

Turning project-specific cleaning code into standardized organization-wide ado-files.

11.1.1 The 'Package' Mindset: Versioning Internal Tools

Why treating your internal scripts like software leads to more sustainable research teams.

11.2 Writing Help Files (.sthlp)

12.1 Introduction to Mata for Evaluators

Matrix manipulation for large-scale administrative data.

12.1.1 The Trifecta Workflow: Stata, Mata, and Python

Strategies for using each language for its specific strength: Stata for management, Mata for computation, and Python for external integration.

Ado-file Idea: `mata_bench`

Proposed Program: `mata_bench`. A utility to benchmark and compare the execution speed of standard Stata loops vs. a Mata implementation of the same logic.

APPENDICES

Appendix A: The Gemini-Stata Setup Guide

Appendix B: The Modern Causal Toolbox

